

MEGHAVI SMART KIOSK

Master Project Documentation & Final Handover

Project Developer: Lokesh Vijjapu

Status: Completed & Deployed

1. Project Overview

Project Name: Meghavi Smart Kiosk

Project Status: Completed & Deployed

Deployment Location: Mumbai Airport

Number of Units Deployed: 3

Platform: Windows 10 (64-bit)

System Type: Interactive Spa Service Kiosk

The Meghavi Smart Kiosk is a fully automated, self-service interactive system designed for spa outlets in high-footfall public locations. The system attracts customers through video content, detects user presence using a camera, and guides users through spa service selection and ordering. Upon successful order placement, a bill is printed automatically, and the system resets to serve the next customer.

The kiosk is designed for continuous 24/7 operation, minimal user intervention, and fast recovery from abnormal conditions.

2. System Architecture (High-Level)

The system consists of the following logical modules:

- Video Playback Module (Idle State)
- Face Detection Module
- Web-Based Spa Application
- Thermal Printer Integration
- Inactivity & Timeout Controller
- Auto-start & Recovery Services

All modules are initialized automatically at system boot.

3. Hardware Configuration

Component	Specification
CPU	Intel i7 – 4th Generation
Display	32-inch TV
Camera	1080p USB Webcam
Printer	KP347 Thermal Printer (Auto-powered)
Network	LAN (Ethernet)
Power Supply	450W SMPS
Operating System	Windows 10 (64-bit)

4. Software Configuration

Operating System: Windows 10 (64-bit)

Application Type: Python-based kiosk controller + Web UI

Startup Mode: Auto-launch on boot

User Login Required: No

Manual Commands: None

All required software, dependencies, and configurations are pre-installed and validated before deployment.

5. Initialization & Power-On Guide

5.1 Pre-Setup

System is plug-and-play

No software installation required at site

5.2 Power-On Sequence

1. Connect LAN cable to kiosk Ethernet port.

2. Ensure stable internet connectivity.
3. Plug kiosk power cable into electrical outlet.
4. Turn ON the main power switch on the kiosk frame.
5. Turn ON the TV and select the correct HDMI source.
6. Press the CPU power button.
7. Verify the AARVEX logo appears during boot.

Note: If the logo does not appear, power OFF the motherboard, wait for 1 minute, and power it ON again.

6. Printer Setup

KP347 thermal printer powers ON automatically with the system.

No manual power switch is required.

Ensure thermal paper is loaded at all times.

Replace paper roll when empty.

7. System Boot & Auto-Start Behavior

Meghavi kiosk application launches automatically after OS boot.

No keyboard, mouse, or operator interaction is required.

System starts in video playback mode with face detection enabled.

8. Functional Working Flow

8.1 Idle State – Video Playback

- Promotional video plays continuously.
- The face detection module runs in the background.

8.2 Face Detection Event

- Video playback pauses.
- The spa website home page opens automatically.
- Face detection remains active.

8.3 User Order Flow

- The user clicks the order button from the home page.
- Order page opens.

- The user selects spa services and confirms order.
- The thermal printer prints bills automatically.
- System redirects back to the spa home page.

8.4 Timeout & Inactivity Handling

- 5 minutes inactivity: Redirects to spa home page.
- 5 seconds no face detected: Returns to video playback mode.

9. Working Principle Summary

- Face detection runs only on video playback and home page.
- Order pages are excluded from face detection to optimize performance.
- System always returns to idle state after order completion or inactivity.
- Designed for unattended public usage.

10. Onsite Deployment & Field Troubleshooting (Mumbai Airport)

10.1 Deployment Summary

- Successfully deployed 3 Meghavi kiosks at Mumbai Airport.
- Performed onsite installation, validation, and live testing independently.

10.2 Issue Observed During Deployment

During onsite verification, one kiosk unit failed to power ON. Initial inspection indicated that the issue occurred due to hardware disturbance during transportation, which is common in long-distance shipping.

10.3 Troubleshooting Actions Performed

The following structured troubleshooting steps were executed onsite:

- Verified external power input and main power switch.
- Inspected SMPS output voltages and power rails.
- Checked motherboard power indicators and connections.
- Identified BIOS-level initialization issue likely caused by CMOS state corruption due to transit shock.

10.4 Resolution

- Performed CMOS reset by shorting the CMOS clear pins for approximately 10 seconds.
- Re-powered the motherboard after reset.

- System booted successfully and resumed normal operation.

10.5 Outcome

- No components were replaced.
- No external support was required.
- Issue was resolved completely onsite.
- System passed post-recovery functional testing and remained stable

This intervention ensured zero hardware replacement cost and prevented deployment delays.

11. Troubleshooting Guide (Operational)

Face Not Detected

- Ensure user is within 100 cm of camera.
- Check lighting conditions.
- Restart kiosk if required.
- Verify internet connectivity.

Spa Website Not Loading

- Check LAN cable connection.
- Confirm internet access.
- Restart system.

Bill Not Printing

- Ensure printer paper is loaded.
- Restart kiosk.

Video Not Playing

- Verify correct HDMI input on TV.
- Check AARVEX logo during boot.
- Restart system if required.

12. Code Reference

- Code is maintained in the internal repository.
- Application auto-starts via system startup configuration.

- Configuration files control: Timeouts, Website URLs, Detection thresholds
- Full source code is intentionally excluded from this document as per best handover practices.

13. Handover Notes

- System is fully automated and stable.
- No daily maintenance required.
- Avoid changing OS or BIOS settings.
- Do not exit kiosk application manually.
- Any updates should be tested offline before deployment.

14. Project Status Summary

Item	Status
Development	Completed
Testing	Completed
Deployment	Completed
Field Validation	Completed
Stability	Verified
Handover Ready	Yes

FINAL STATEMENT

The Meghavi Smart Kiosk project has been successfully developed, deployed, and validated onsite. The system is production-ready and can be maintained by a new engineer using this document without dependency on the original developer.